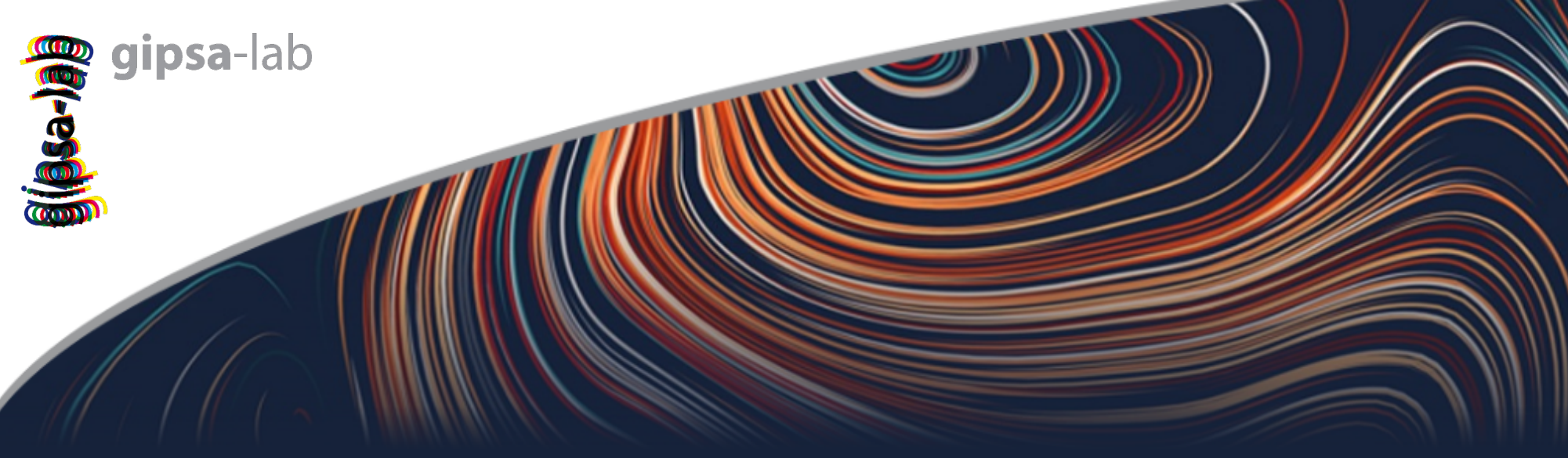


Optimal sensor placement for source separation with application to event-related potential estimation

PhD Defense, by Mohammad SADEGHI

Supervisors: Bertrand RIVET, Massoud BABAIE-ZADEH

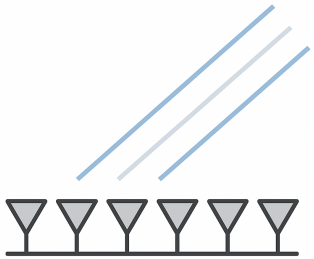
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30th June 2026

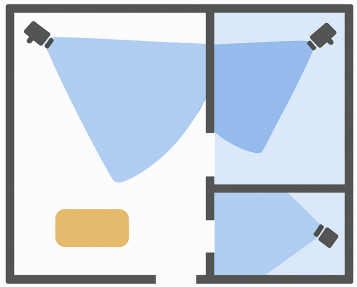
Introduction

Sensing as the Foundation of Modern Systems

MIMO radar



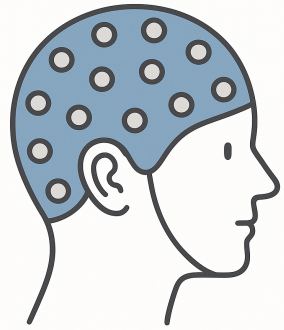
CCTV cameras



Microphone



EEG cap



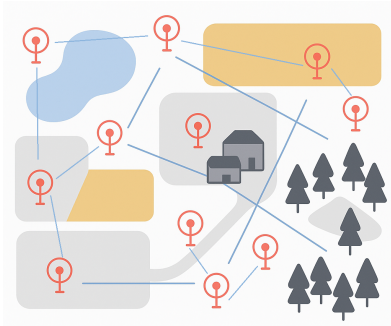
Smartphone



Car sensors



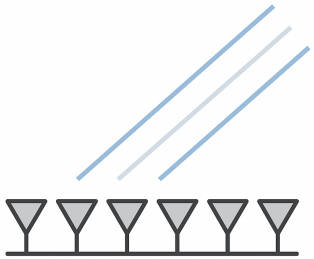
Environmental monitoring



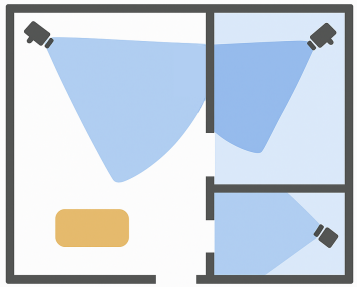
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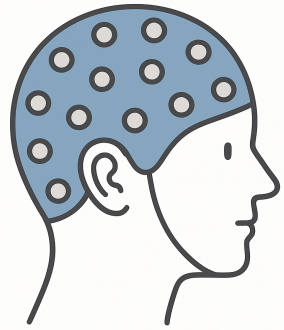
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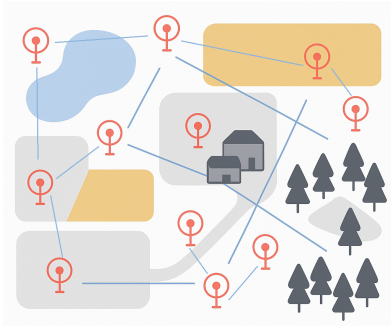
Smartphone



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Where to Place the Sensors ?

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Optimal Sensor Placement

More sensors lead to :

- ↑ **More Information** (Richer measurements, better coverage)

Optimal Sensor Placement

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- ↑ **System Cost** (Hardware, Maintenance)
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Optimal Sensor Placement

More sensors lead to :

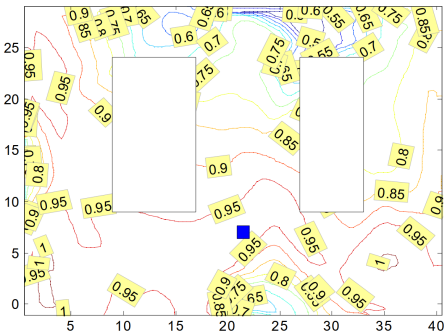
- ↑ **More Information** (Richer measurements, better coverage)
- ↑ **System Cost** (Hardware, Maintenance)
- ↑ **Computational Complexity** (Processing, Storage)
- ↓ **User Comfort** (e.g., EEG caps, wearables)

Limited number of sensors $\Longrightarrow$ Optimal placement is critical

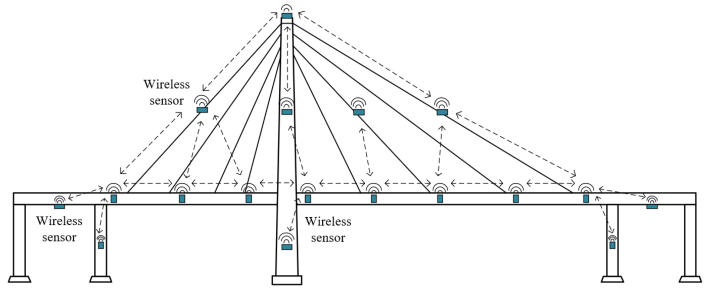
Examples of Optimal Sensor Placement

Optimality is task-dependent :

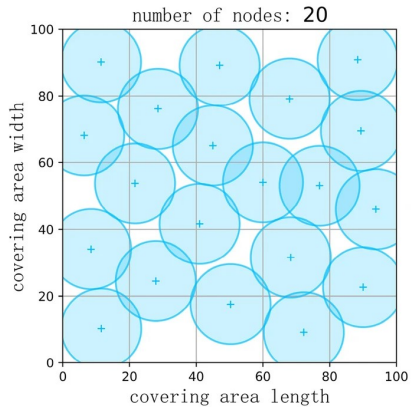
- ↑ **Information gain**
- ↓ Remaining **uncertainty**
- ↑ **Estimation accuracy** of unknown parameters
- ↑ **Coverage** of the target area



Temperature measurement,
(Krause et al. 2008)



Structural health monitoring,
(Deng et al. 2023)



Coverage in Wireless sensor network, (Yang et al. 2025)

Bayesian Experimental Design Criteria

Sensor placement for a random field (e.g., GP)

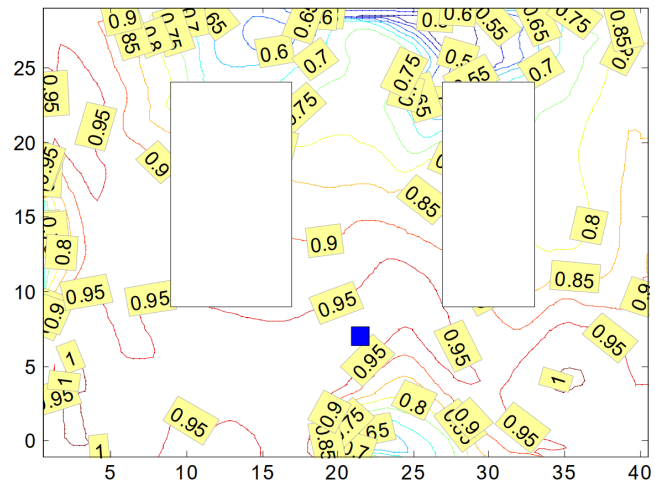
Two classical sensor placement criteria for accurate estimating a phenomenon (e.g., temperature) modeled as a random field.

Entropy Criterion (Shewry et al. 1987)

idea : Place sensors where the measurements are most informative.
$\rightarrow$ Maximize entropy at the chosen sensor locations.

Mutual Information Cri. (Krause et al. 2008)

idea : Place sensors where maximally reduce uncertainty over the entire field.
$\rightarrow$ Maximize mutual information between placed & remaining locations.

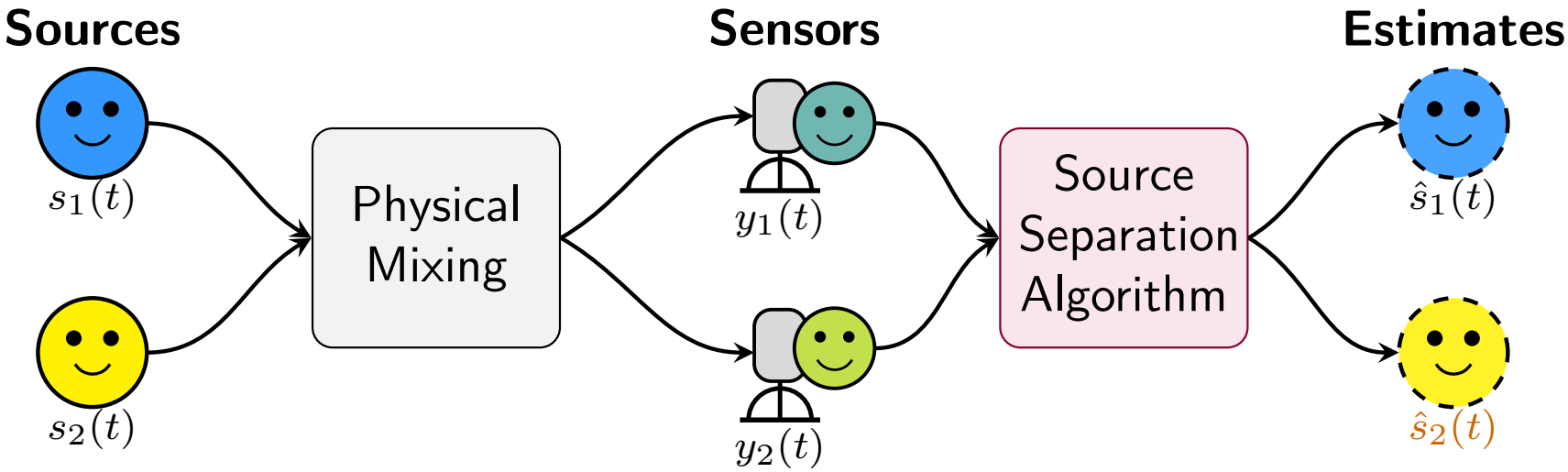


Temperature measurement
(Krause et al. 2008)

The Target Application : Source Separation

What if sensors measure a mixture of signals ?

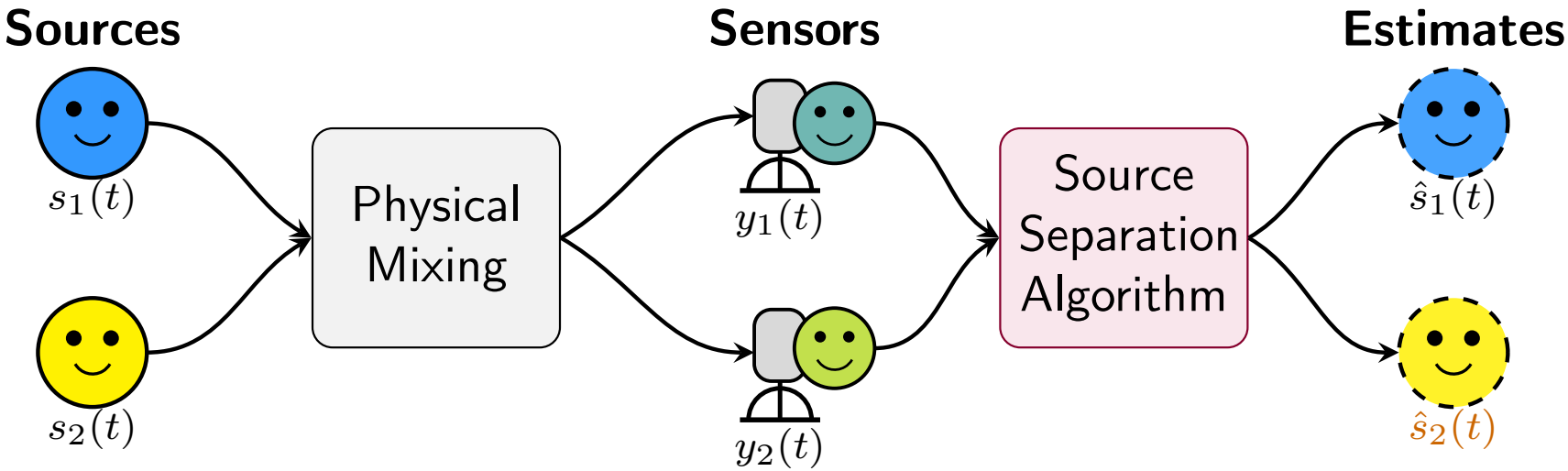
- Each sensor measures a **mixture** of underlying **source signals**
- Goal : **recover** the original sources from sensor measurements



The Target Application : Source Separation

What if sensors measure a mixture of signals ?

- Each sensor measures a **mixture** of underlying **source signals**
- Goal : **recover** the original sources from sensor measurements



This thesis: Optimal **sensor placement** to maximise source separation quality.

Introduction

Optimal Sensor Placement for Source Separation Under Noise

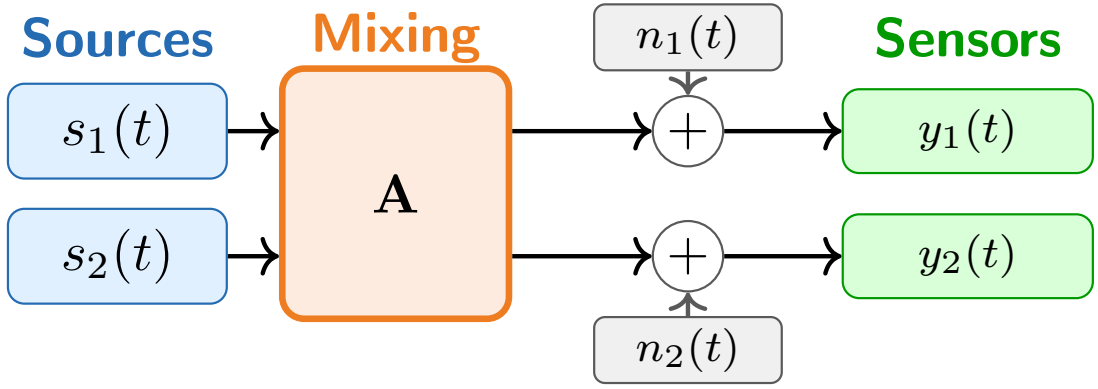
Application in P300-based brain computer interface

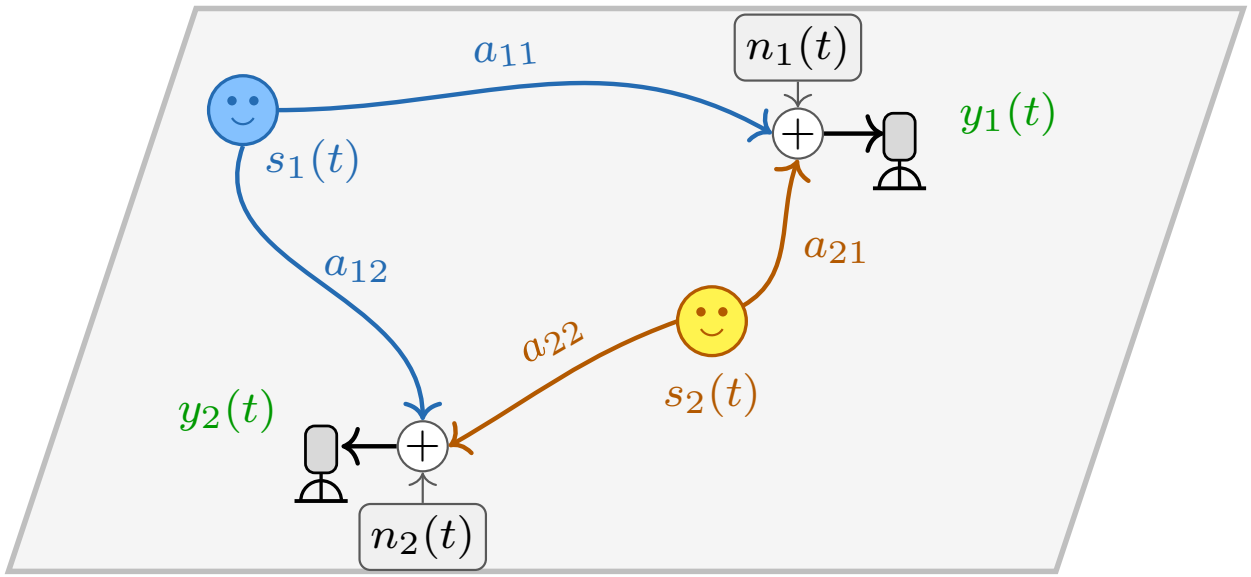
Sensor placement for sparse reconstruction

Conclusion

Linear Mixing Model

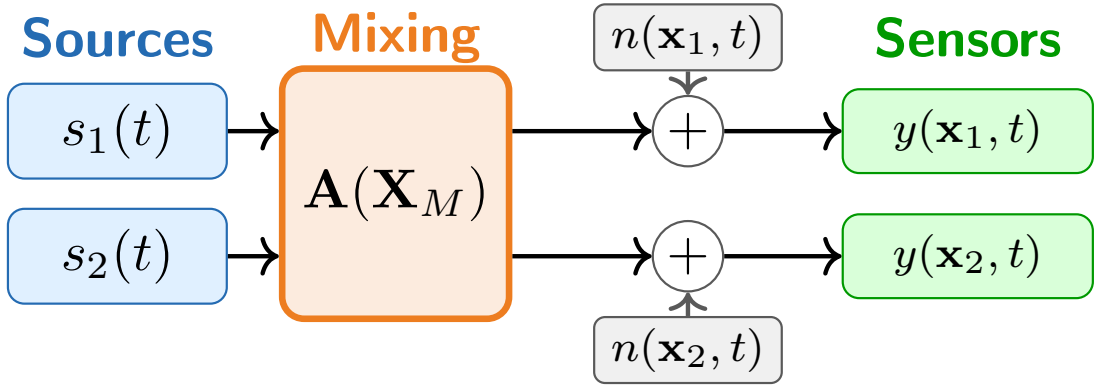
A simple example

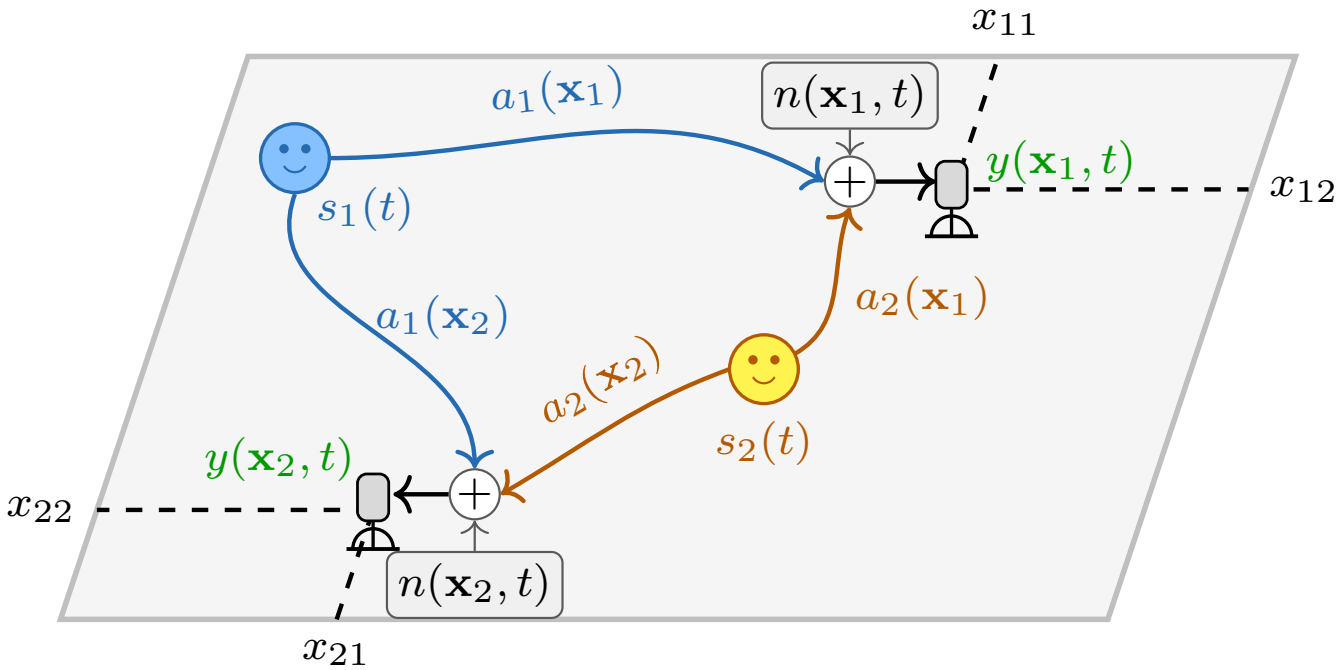




Linear Mixing Model

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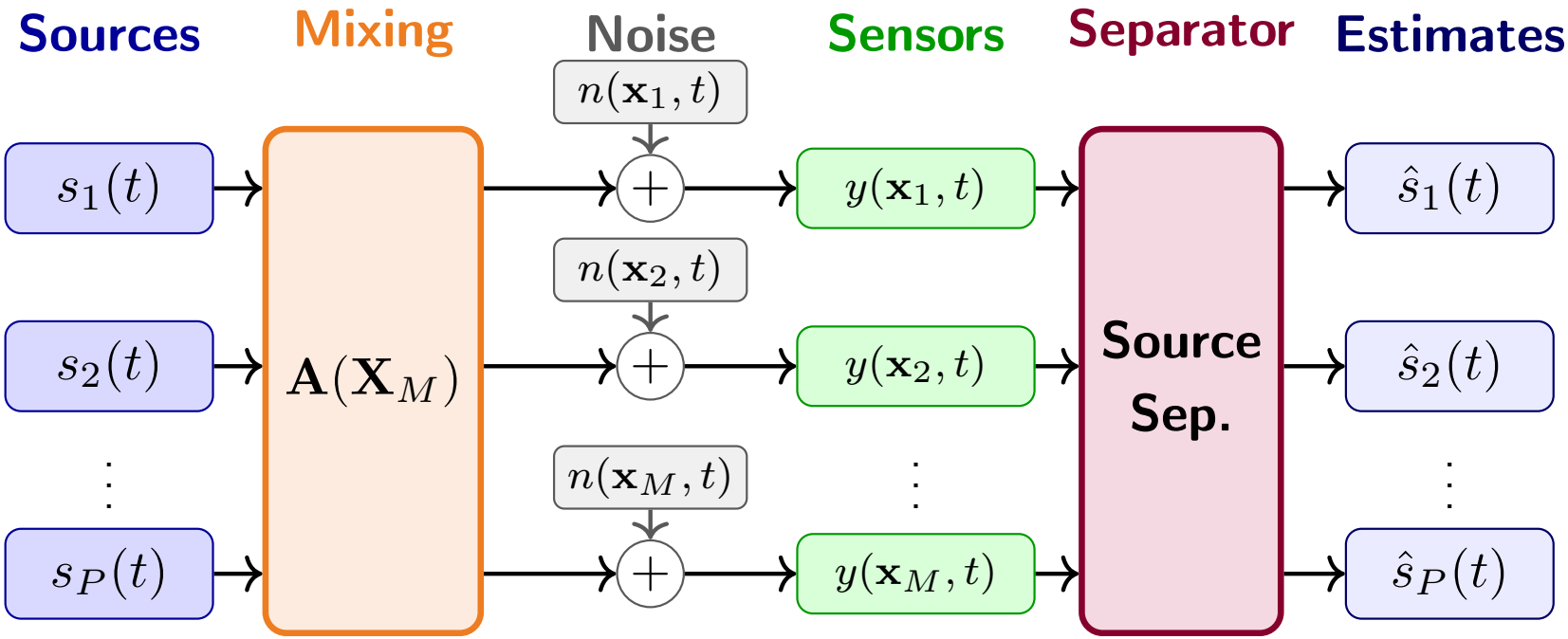




Source Separation

Mixing, noise, and the role of sensor placement

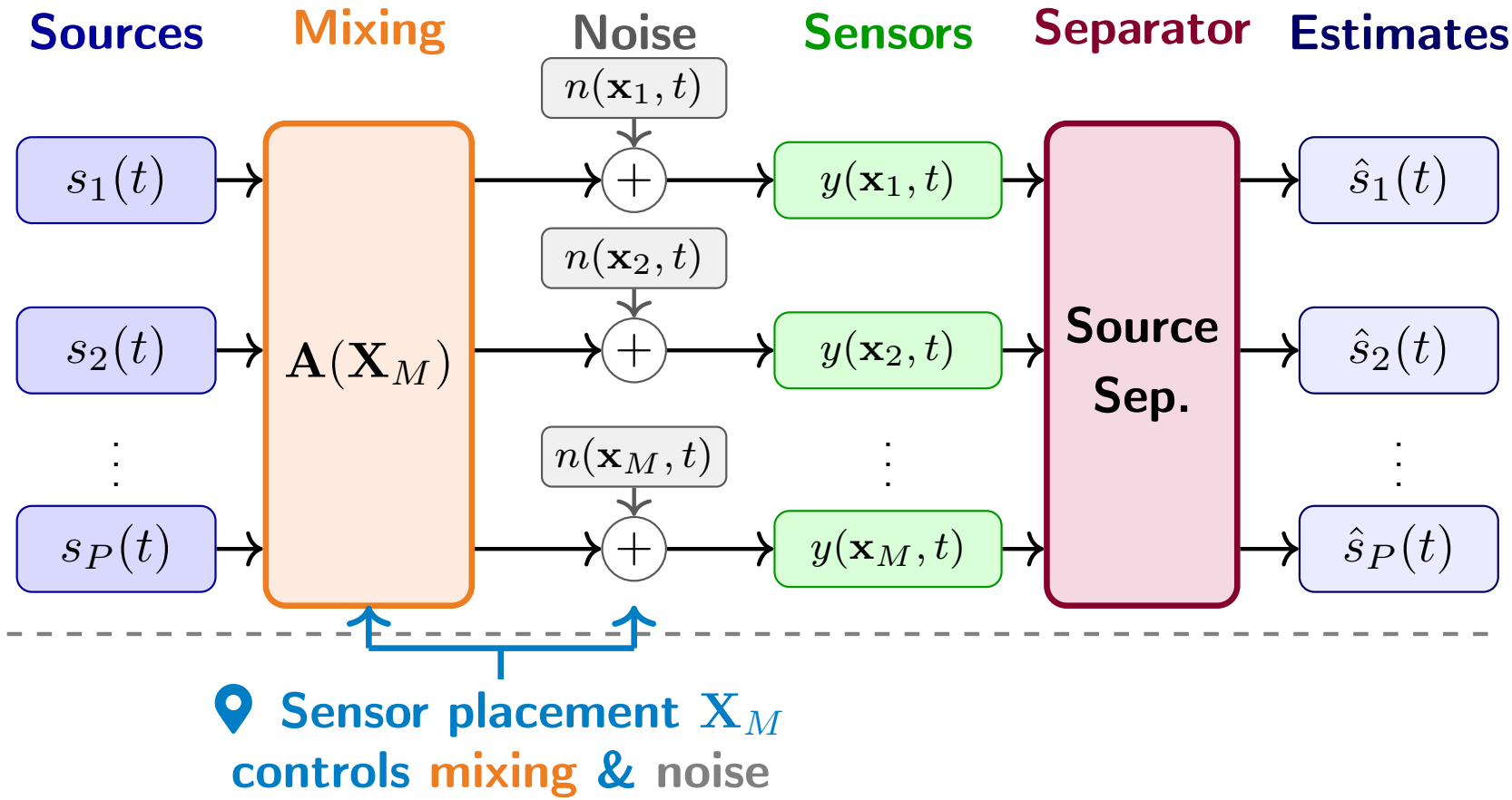
- Coordinates of the sensor m : $\mathbf{x}_m \in \mathcal{X} \subset \mathbb{R}^D$
- All M locations stacked as $\mathbf{X}_M \triangleq [\mathbf{x}_1, \ldots, \mathbf{x}_M]$



Source Separation

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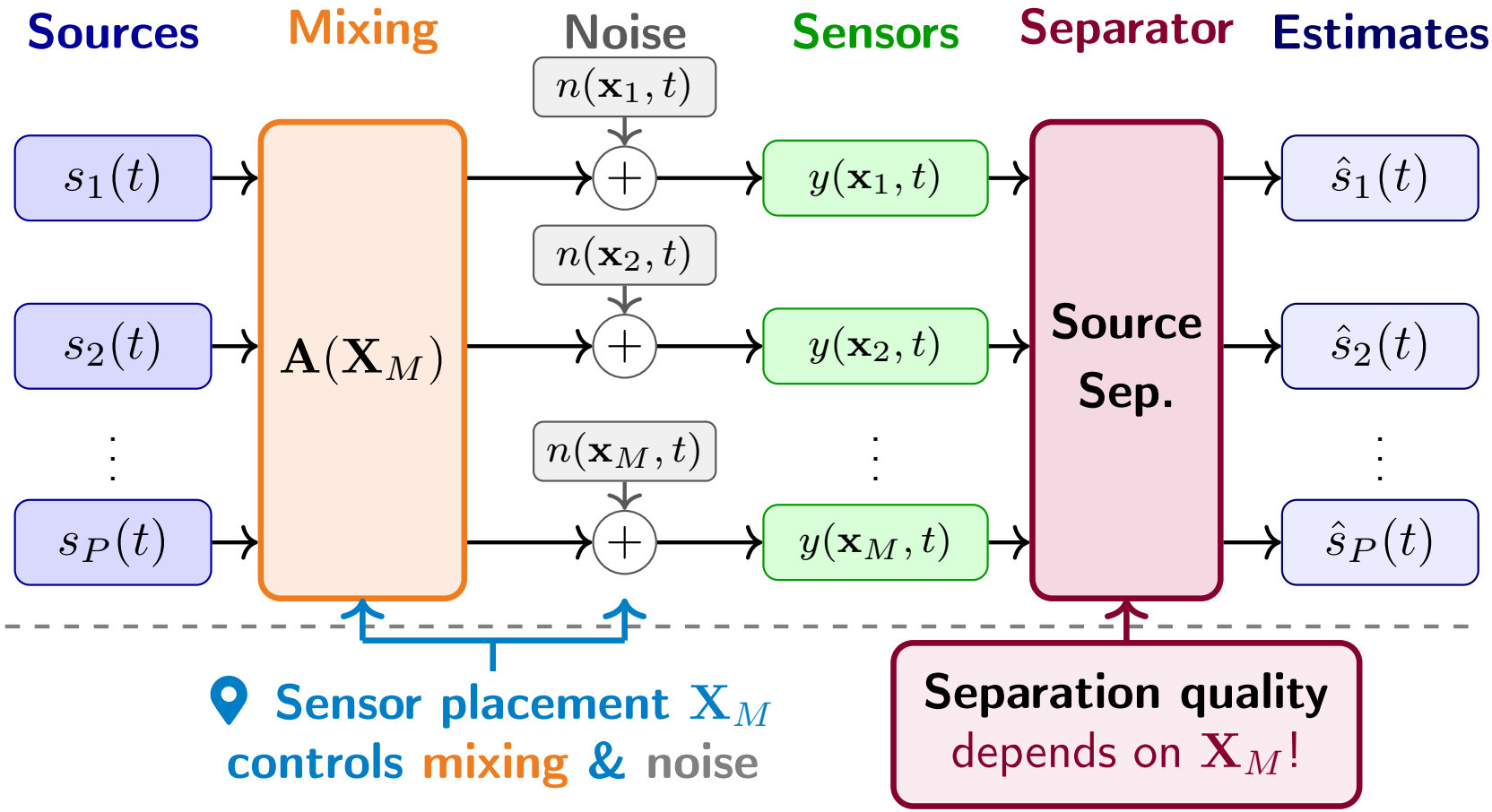
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Source Separation

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Linear Mixing Model

- Spatial gain from source p to a sensor located at $\mathbf{x}_m$: $\boxed{a_p(\mathbf{x}_m)}$
- Measured signal by sensor m :

$$y(\mathbf{x}_m, t) = \sum_{p=1}^{P} a_p(\mathbf{x}_m)\, s_p(t) + n(\mathbf{x}_m, t)$$

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$$y(\mathbf{x}_m, t) = \sum_{p=1}^{P} a_p(\mathbf{x}_m)\, s_p(t) + n(\mathbf{x}_m, t)$$

- Stacking measurements and spatial gains in vectors :

$$\begin{aligned}\mathbf{y}(\mathbf{X}_M, t) &= \sum_{p=1}^{P} \mathbf{a}_p(\mathbf{X}_M)\, s_p(t) + \mathbf{n}(\mathbf{X}_M, t) \\ &= \mathbf{A}(\mathbf{X}_M)\, \mathbf{s}(t) + \mathbf{n}(\mathbf{X}_M, t)\end{aligned}$$

$$\begin{aligned}\mathbf{a}_p(\mathbf{X}_M) &= [a_p(\mathbf{x}_1), \ldots, a_p(\mathbf{x}_M)]^T \quad , \quad \mathbf{A}(\mathbf{X}_M) = [\mathbf{a}_1(\mathbf{X}_M), \ldots, \mathbf{a}_P(\mathbf{X}_M)] \\ \mathbf{y}(\mathbf{X}_M, t) &= [y(\mathbf{x}_1, t), \ldots, y(\mathbf{x}_M, t)]^T \quad , \quad \mathbf{s}(t) = [s_1(t), s_2(t), \ldots, s_P(t)]^T\end{aligned}$$

Proposed placement criteria

Signal-to-interference-plus-noise ratio (SINR)

Linear estimation of the l-th source using a vector $\mathbf{f}_l \in \mathbb{R}^M$:

$$\hat{s}_l(t) = \mathbf{f}_l^T \mathbf{y}(\mathbf{X}_M, t) = \sum_{p=1}^{P} \mathbf{f}_l^T \mathbf{a}_p(\mathbf{X}_M) s_p(t) + \mathbf{f}_l^T \mathbf{n}(\mathbf{X}_M, t)$$

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Objective : Maximize the SINR.

$$\text{SINR}_l(\mathbf{f}_l; \mathbf{X}_M) = \frac{\mathbf{f}_l^T \mathbf{a}_l(\mathbf{X}_M) \mathbf{a}_l(\mathbf{X}_M)^T \mathbf{f}_l}{\mathbf{f}_l^T \left(\sum_{p=1, p \neq l}^{P} \mathbf{a}_p(\mathbf{X}_M) \mathbf{a}_p(\mathbf{X}_M)^T + \mathbf{C}^n(\mathbf{X}_M) \right) \mathbf{f}_l}$$

Assumptions :

- The sources : independent with zero means and unit variances.
- The noise covariance matrix : $\mathbf{C}^n(\mathbf{X}_M)$.

Proposed placement criteria

Signal-to-interference-plus-noise ratio (SINR)

Optimal filter $\mathbf{f}_l^*$ (Van Trees et al. 2002) :

$$\mathbf{f}_l^*(\mathbf{X}_M) = \Big(\sum_{\substack{p=1 \\ p \neq l}}^{P} \mathbf{a}_p(\mathbf{X}_M)\mathbf{a}_p(\mathbf{X}_M)^\top + \mathbf{C}^n(\mathbf{X}_M) \Big)^{-1} \mathbf{a}_l(\mathbf{X}_M)$$

Resulting maximum SINR :

$$\mathrm{SINR}_l(\mathbf{f}_l^*, \mathbf{X}_M) = \mathbf{a}_l(\mathbf{X}_M)^T \Big(\sum_{\substack{p=1 \\ p \neq l}}^{P} \mathbf{a}_p(\mathbf{X}_M)\mathbf{a}_p(\mathbf{X}_M)^T + \mathbf{C}^n(\mathbf{X}_M) \Big)^{-1} \mathbf{a}_l(\mathbf{X}_M)$$

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Placement criterion : Sum of SINRs over sources of interest

$$J_{\mathrm{SINR}}(\mathbf{X}_M) = \sum_{l \in \mathcal{L}} \mathrm{SINR}_l(\mathbf{f}_l^*, \mathbf{X}_M) \quad , \quad \mathcal{L} \subset \{1, 2, ..., P\}$$

Prior Knowledge on Spatial Gains

Why a Stochastic Model ?

$$J_{\mathrm{SINR}}(\mathbf{X}_M) = \sum_{l \in \mathcal{L}} \mathbf{a}_l(\mathbf{X}_M)^T \left(\sum_{\substack{p=1 \\ p \neq l}}^{P} \mathbf{a}_p(\mathbf{X}_M)\mathbf{a}_p(\mathbf{X}_M)^T + \mathbf{C}^n(\mathbf{X}_M) \right)^{-1} \mathbf{a}_l(\mathbf{X}_M)$$

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Needs exact spatial gain functions $a_p(\mathbf{x})$.

- $a_p(\mathbf{x})$: not directly measurable — estimated from placed sensors measurements.
- Gains may change between training and deployment (e.g., EEG applications).

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Solution

Model $a_p(\cdot)$ with uncertainty $\Rightarrow$ Stochastic modeling

Prior Knowledge on Spatial Gains

Gaussian Process

Each source's spatial gain : a draw from a **Gaussian Process**,

$$a_p(\cdot) \sim \mathcal{GP}(m^{a_p}(\cdot),\ C^{a_p}(\cdot,\cdot))$$

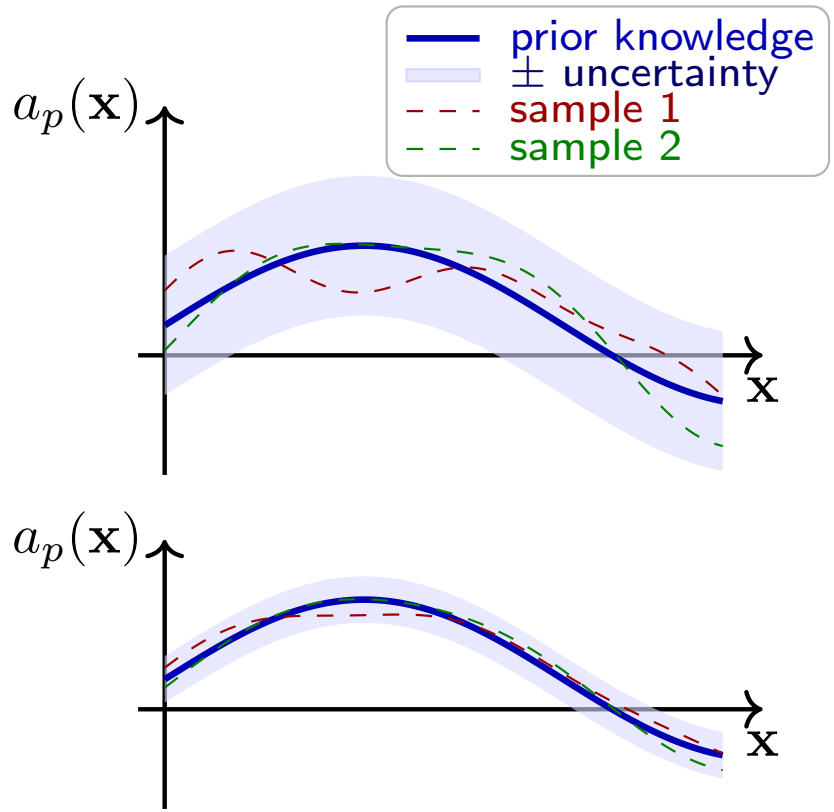
Mean function $m^{a_p}(\mathbf{x})$:

- Best available prior knowledge

Covariance function $C^{a_p}(\mathbf{x}, \mathbf{x}')$:

- Models uncertainty and spatial correlation
- Example : Squared exponential kernel,

$$C^{a_p}(\mathbf{x}, \mathbf{x}') = \sigma_p^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\rho_p^2}\right)$$



Expected Criterion

J_{SINR} depends on stochastic spatial gains $\implies$ it is stochastic.

$$\hat{J}_{\mathrm{SINR}}(\mathbf{X}_M) = \mathbb{E}\left[J_{\mathrm{SINR}}(\mathbf{X}_M)\right]$$

$$= \sum_{l \in \mathcal{L}} \mathbb{E}\left[\mathbf{a}_l(\mathbf{X}_M)^T \left(\sum_{\substack{p=1 \\ p \neq l}}^{P} \mathbf{a}_p(\mathbf{X}_M)\mathbf{a}_p(\mathbf{X}_M)^T + \mathbf{C}^n(\mathbf{X}_M) \right)^{-1} \mathbf{a}_l(\mathbf{X}_M)\right]$$

- No closed-form expectation available.
- Monte Carlo averaging $\rightarrow$ criterion & gradient evaluation.

Proposed placement criteria

Mean Squared Error (MSE)

LMMSE estimator : $\hat{\mathbf{s}}(t) = \mathbf{B}\mathbf{y}(\mathbf{X}_M, t)$ (Comon et al. 2010)

$$\begin{aligned}\mathbf{B}^* =& \underset{\mathbf{B}}{\operatorname{argmin}}\, \mathbb{E}\left\{\|\mathbf{s}(t) - \mathbf{B}\mathbf{y}(\mathbf{X}_M, t)\|_2^2\right\} \\ =& \mathbf{A}^T\left(\mathbf{A}\mathbf{A}^T + \mathbf{C}^n\right)^{-1}\end{aligned}$$

MSE :

$$\begin{aligned}&\mathbb{E}\left\{\|\mathbf{s}(t) - \mathbf{B}^*\mathbf{y}(\mathbf{X}_M, t)\|^2\right\} = \\ &\qquad P - \operatorname{Tr}\left(\mathbf{A}^T(\mathbf{A}\mathbf{A}^T + \mathbf{C}^n)^{-1}\mathbf{A}\right)\end{aligned}$$

MSE criterion :

$$\boxed{\hat{J}_{\mathrm{MSE}}(\mathbf{X}_M) = \mathbb{E}\left\{\operatorname{Tr}\left(\mathbf{A}^T(\mathbf{A}\mathbf{A}^T + \mathbf{C}^n)^{-1}\mathbf{A}\right)\right\}}$$

Solving the Optimization Problem

High-dimensional $(M \times D)$ non-convex optimization :

$$\mathbf{X}_M^* = \underset{\mathbf{x}_1,\ldots,\mathbf{x}_M \in \mathcal{D}}{\operatorname{argmax}} \; J(\mathbf{X}_M) \begin{cases} \nearrow \text{SINR} \\ \searrow \text{MSE} \end{cases}$$

Direct solution difficult $\rightarrow$ high computational complexity.

Solving the Optimization Problem

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Three approaches to render the problem tractable :

Greedy Method

Step by step optimization : **one** sensor at each step, keeping others fixed

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Three approaches to render the problem tractable :

Greedy Method	**Sequential Approach**
Step by step optimization : **one** sensor at each step, keeping others fixed	In each greedy step, update GP model via measurements of placed sensors

Solving the Optimization Problem

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Direct solution difficult $\rightarrow$ high computational complexity.

Three approaches to render the problem tractable :

Greedy Method	**Sequential Approach**	**Sensor Refinement**
Step by step optimization : **one** sensor at each step, keeping others fixed	In each greedy step, update GP model via measurements of placed sensors	Start from initial locations, simultaneous gradient update of sensor locations

Greedy Method

Break down M-sensor optimization problem into M sub-problems :

$$\mathbf{X}_0^* = \{\}$$
$$\textbf{for } k = 1 : M$$
$$\quad \mathbf{x}_k^* = \underset{\mathbf{x}_k \in \mathcal{D}}{\operatorname{argmax}}\ J(\mathbf{X}_{k-1}^*, \mathbf{x}_k) \quad (*)$$
$$\quad \mathbf{X}_k^* = \{\mathbf{X}_{k-1}^*, \mathbf{x}_k^*\}$$
$$\textbf{end}$$

$(*)$ Computationally feasible : optimize over $\mathbf{x}_k \in \mathbb{R}^D$ only.

Two methods to solve $(*)$:

- **Grid search** — evaluate J on a dense grid ; inefficient.
- **Multistart gradient-based** — start from coarse grid, converge to local optima, pick best $\rightarrow$ efficient & accurate.

Gradient-Based Local Optimization

Local optimization via gradient of the objective function.

Gradient Ascent (GA)

$$\mathbf{x}_k^{(n+1)} = \mathbf{x}_k^{(n)} + \mu\, \mathbb{E}\{\nabla_{\mathbf{x}_k} J(\mathbf{X}_k)\}$$

- Expectation approximated by Monte Carlo averaging.

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Stochastic Gradient Ascent (SGA)

$$\mathbf{x}_k^{(n+1)} = \mathbf{x}_k^{(n)} + \mu \, \nabla_{\mathbf{x}_k} j(\mathbf{X}_k)$$

- $\nabla_{\mathbf{x}_k} j(\mathbf{X}_k)$: single Monte Carlo sample of gradient $\rightarrow$ noisy but computationally cheap.
- Convergence verified numerically.

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Experimental Setup

Simulation Environment

- Area : 2D unit square, $\mathbf{x} \in [0, 1]^2$.
- Sensors M : 1 to 15.
- Monte Carlo runs : 100.

Experiment 1 : Method Comparison

- **Task** : Source Extraction ($P = 5$) & Separation ($P = 3$).
- **Compared methods :**
  - Proposed SINR and MSE criteria + greedy optimization
  - MI and entropy baselines
  - For source extraction : expected SNR and robust SNR

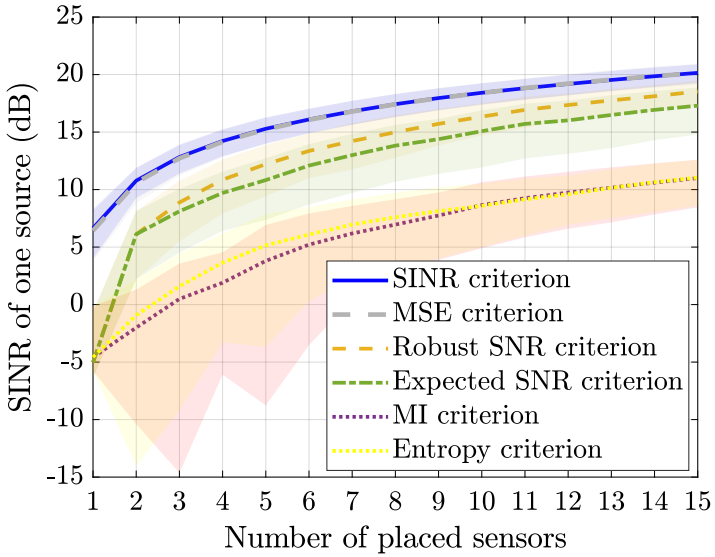
Experiment 2 : Uncertainty Effect

- **Task** : Compare Greedy vs. Sequential approaches.
- **Uncertainty Levels**, low : $\sigma_u = 0.3$, high : $\sigma_u = 0.8$.

Numerical Results

Experiment 1 : Method Comparison

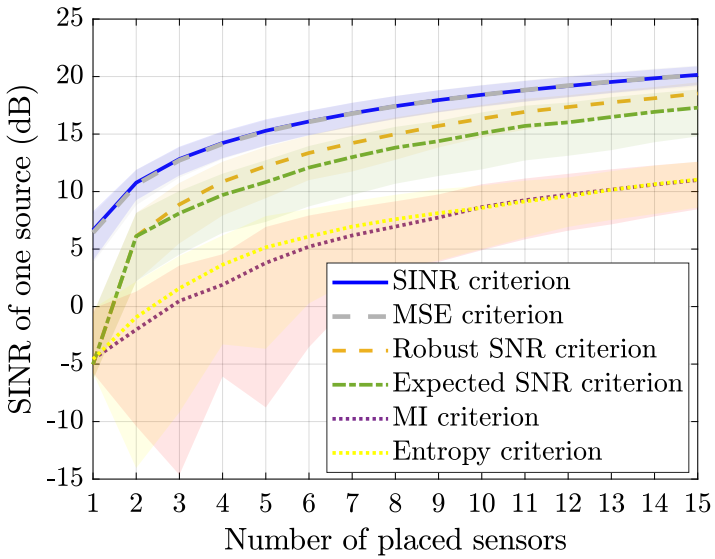
Source Extraction Task



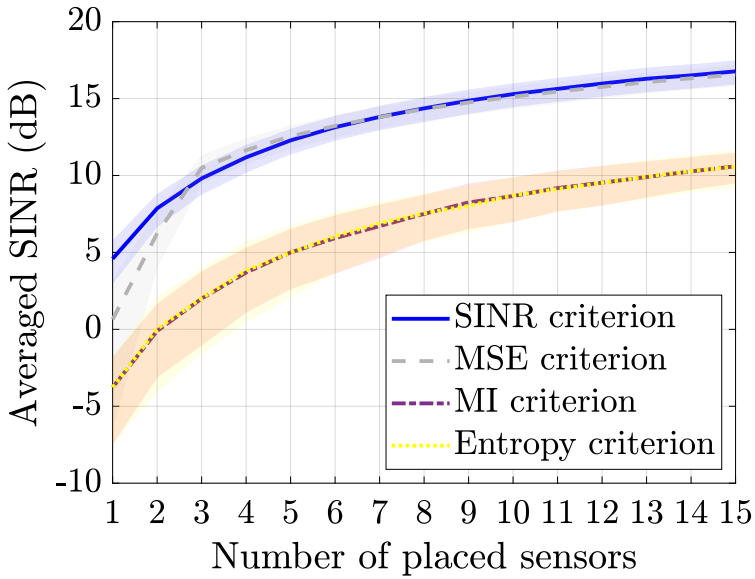
Numerical Results

Experiment 1 : Method Comparison

Source Extraction Task



Source Separation Task



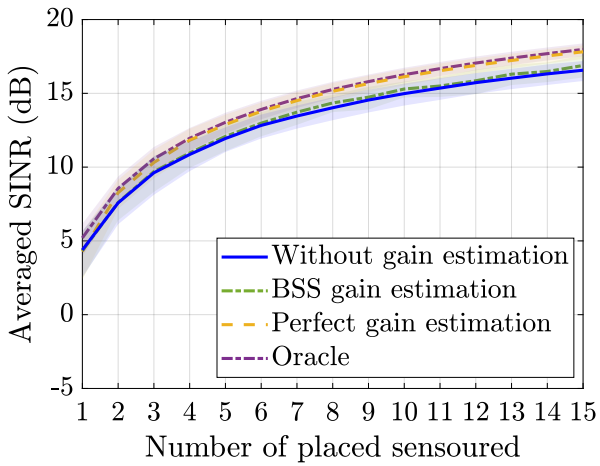
- SINR & MSE criteria outperform Bayesian design & source extraction criteria $\rightarrow$ higher output SINR of estimated sources.

Numerical Results

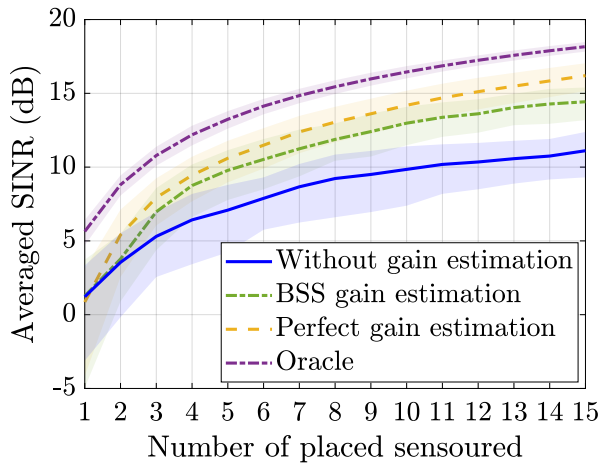
Experiment 2 : Uncertainty Effect

SINR criterion, source separation task, with greedy and sequential optimization methods.

Low uncertainty, $\sigma_u = 0.3$



High uncertainty, $\sigma_u = 0.8$



- Higher level of uncertainty→ Lower performance, higher deviation
- Sequential GP updating improves performance effectively.

Introduction

Optimal Sensor Placement for Source Separation Under Noise

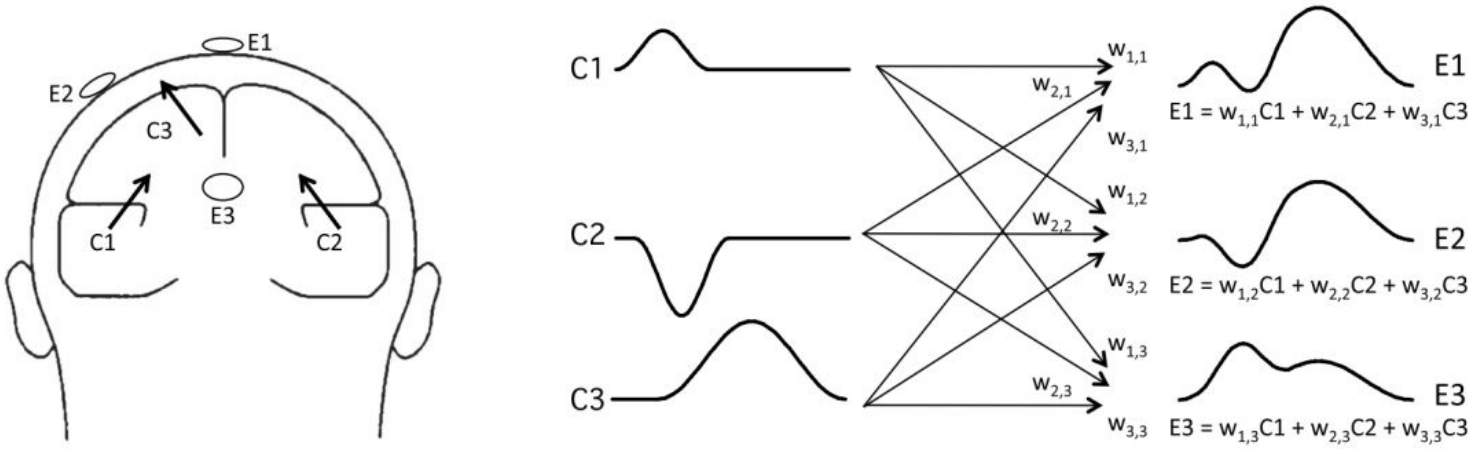
Application in P300-based brain computer interface

Sensor placement for sparse reconstruction

Conclusion

The EEG Forward Model

EEG signals : Can be modeled as instantaneous linear mixing of multiple brain source activities.

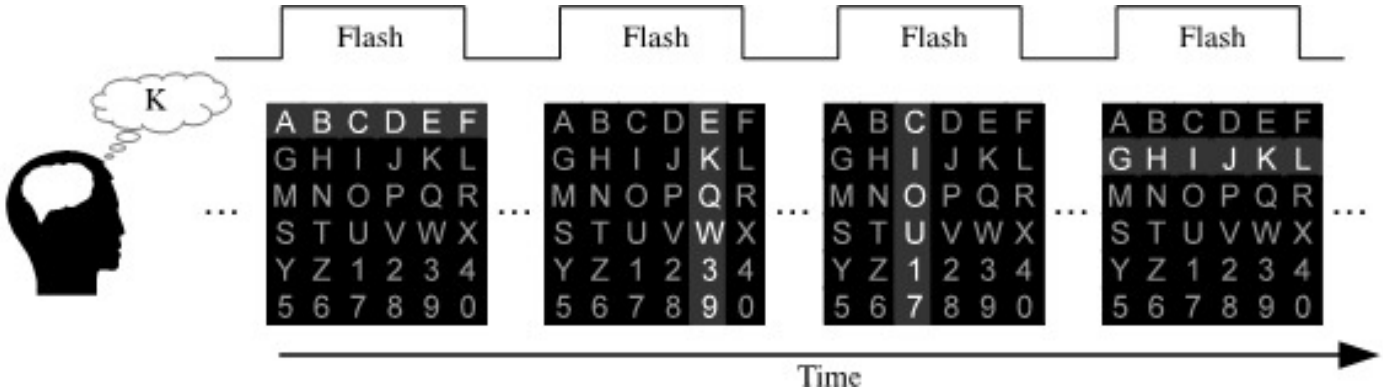


(Luck 2014)

$$\mathbf{y}(\mathbf{X}_M, t) = \mathbf{A}(\mathbf{X}_M)\,\mathbf{s}(t) + \mathbf{n}(\mathbf{X}_M, t)$$

P300-based Brain-Computer Interfaces

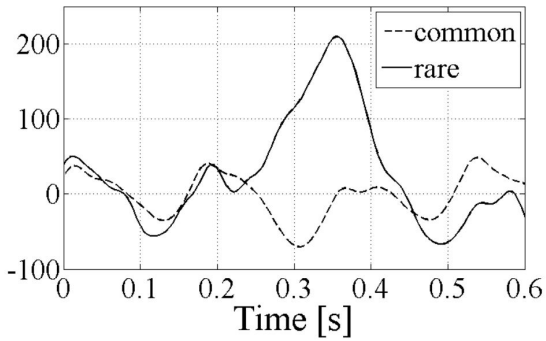
P300 Speller



(Sato et al. 2016)

Goal : Separate P300 sources from EEG recordings for classification.

- P300 : ERP elicited by rare events.

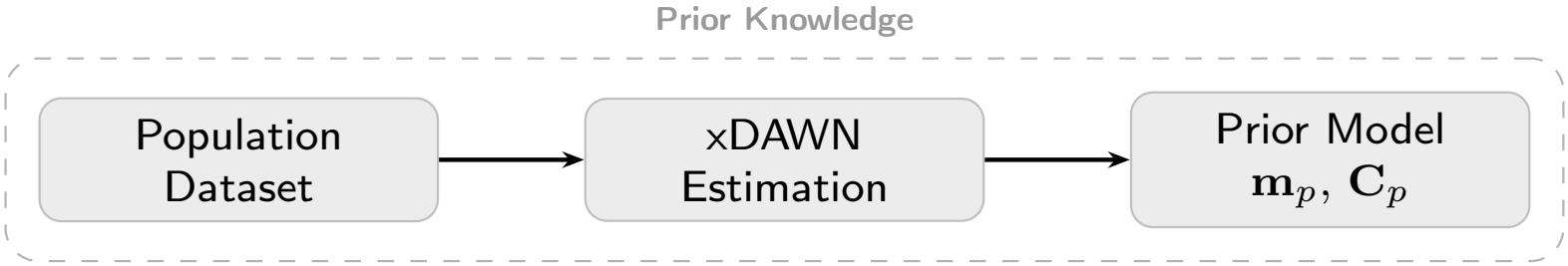


Stochastic Model of Spatial Gains

Learning prior knowledge

Prior knowledge on the spatial gains for P300 related sources :

$$\mathbf{a}_p(\mathbf{X}_{N_s}) \sim \mathcal{N}(\mathbf{m}_p, \mathbf{C}_p)$$

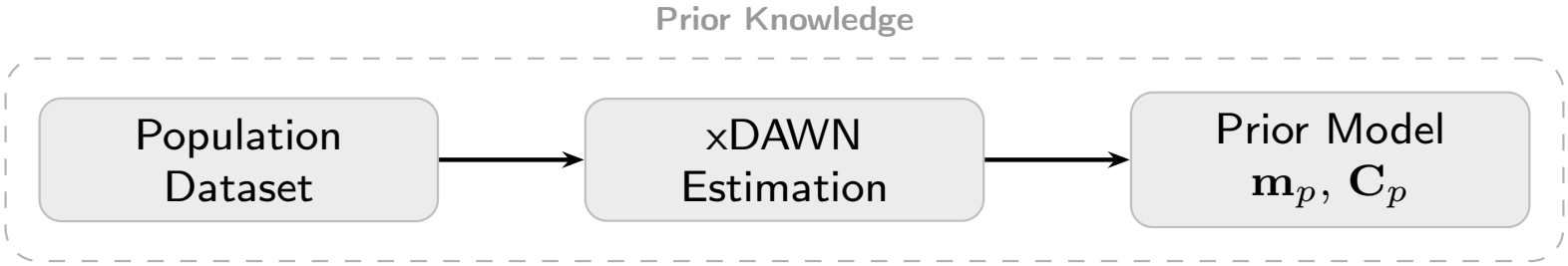


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- For each subject n in training dataset $\xrightarrow{\text{xDAWN}} \mathbf{a}_p^{(n)}(\mathbf{X}_{N_s})$

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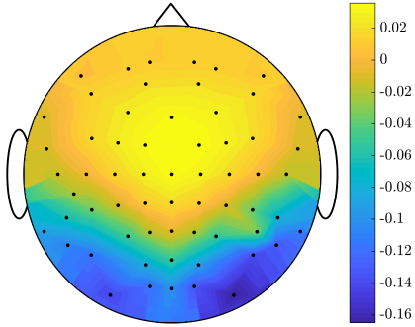
- For each subject n in training dataset $\xrightarrow{\text{xDAWN}} \mathbf{a}_p^{(n)}(\mathbf{X}_{N_s})$
- Learn model parameters using samples provided by subjects :

$$\mathbf{m}_p = 1/T \sum_{n=1}^{T} \mathbf{a}_p^{(n)},$$

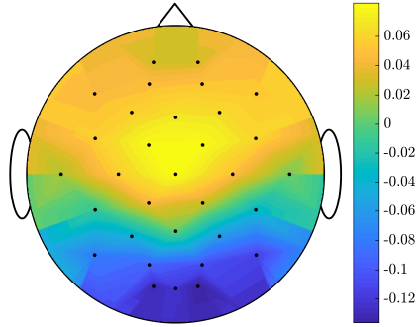
$$\mathbf{C}_p = 1/T \sum_{n=1}^{T} (\mathbf{a}_p^{(n)} - \mathbf{m}_p)^T (\mathbf{a}_p^{(n)} - \mathbf{m}_p)$$

Scalp Distribution of the Mean of Spatial Gain

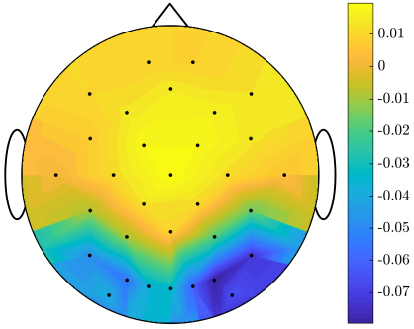
Consistency over different datasets



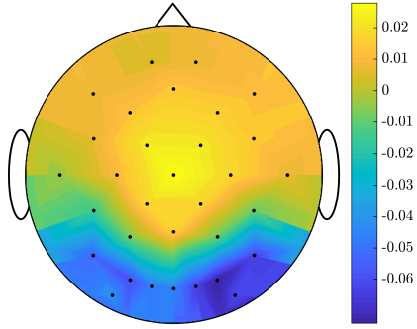
(a) Dataset #1



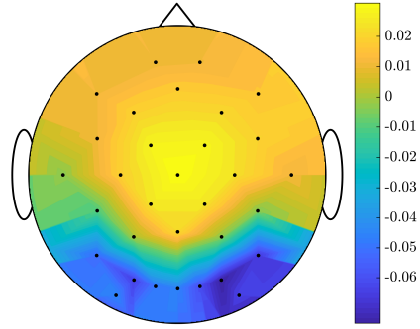
(b) Dataset #2



(c) Dataset #3, Session 1



(d) Dataset #3, Session 2



(e) Dataset #3, Session 3

Experiment

Comparing Subject Independent Electrode Placement Methods

Proposed : Subject Independent Electrode Placement (SIEP)

- Learn prior knowledge from population of subjects
- Use optimized placement (SINR) based on prior for new subject

Subject Dependent Oracle (SDO) : proposed method with perfect $\mathbf{a}_p$

- Subject specific electrode placement knowing full knowledge.

Experiment

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Subject Dependent Oracle (SDO) : proposed method with perfect $\mathbf{a}_p$

- Subject specific electrode placement knowing full knowledge.

Standard Electrode Placement (SEP)

- Place electrodes in fixed standard locations (4, 8 or 16 sensors)

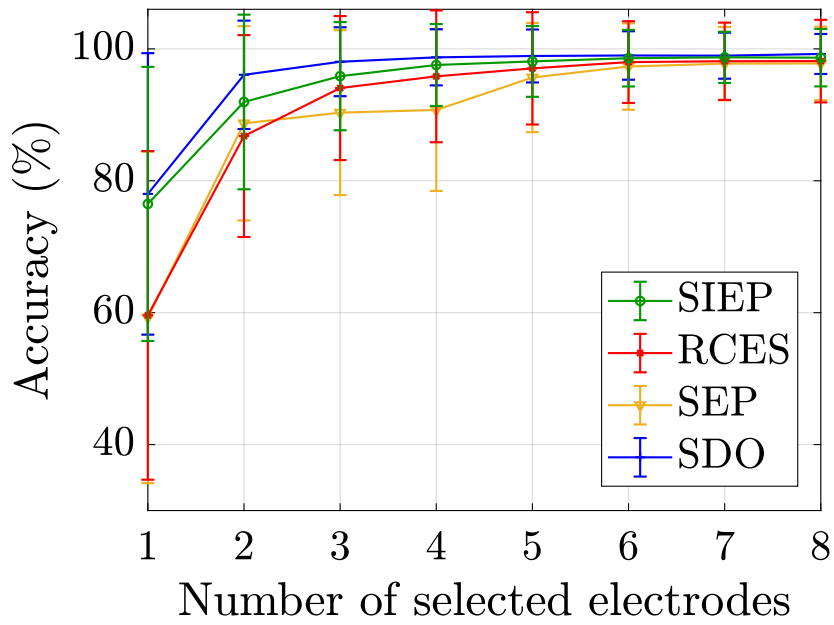
Rank-Based Common Electrode Selection (RCES)

- Find optimal placement for each subject
- Rank electrodes by selection frequency, choose most frequents.

Numerical results

Comparing subject independent placement methods

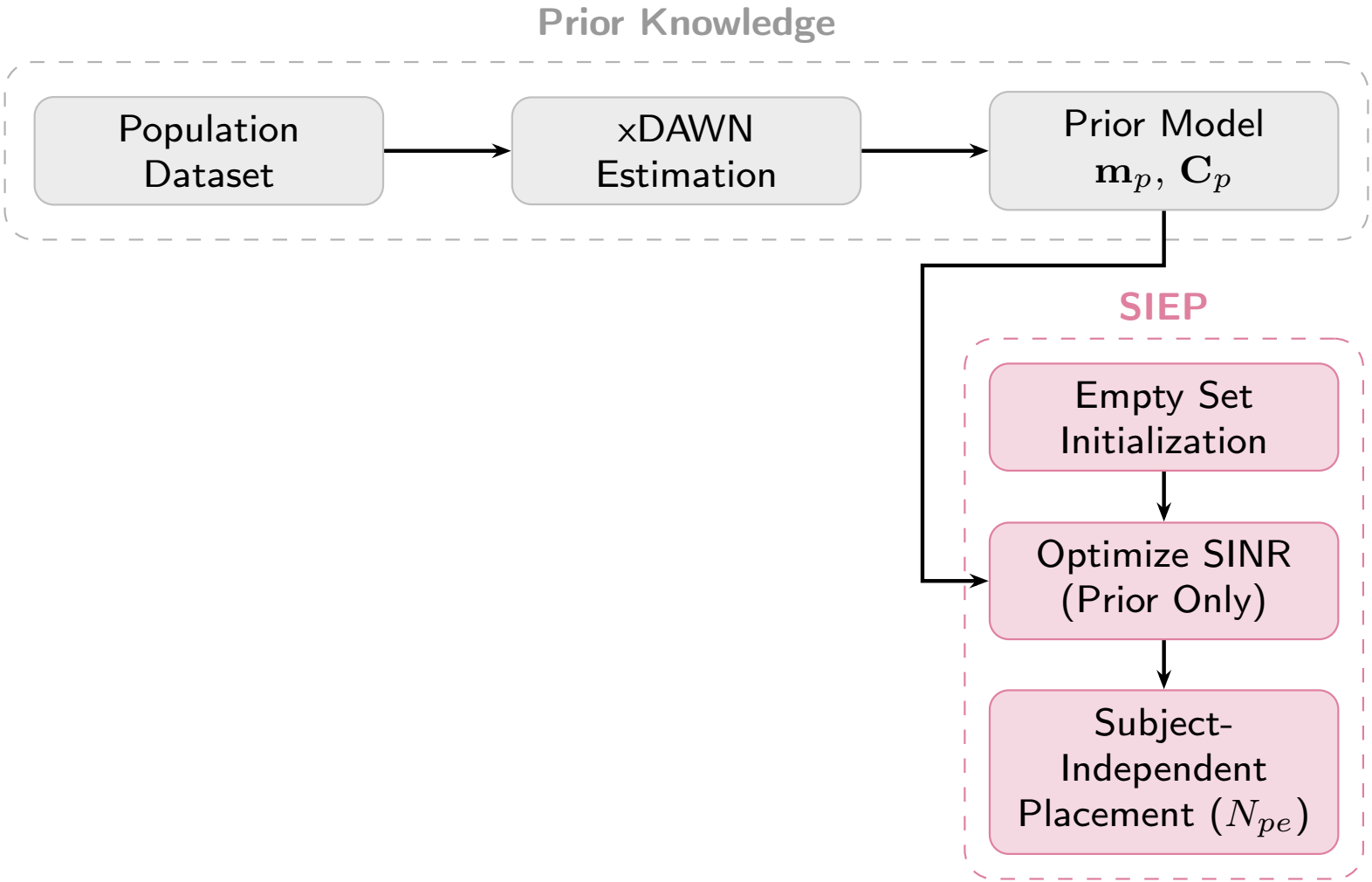
Average accuracy over subjects, Dataset #1, xDAWN-BLDA classifier



- SIEP outperforms standard placement (SEP) for all placed electrode counts, and RCES especially for small electrode sets.

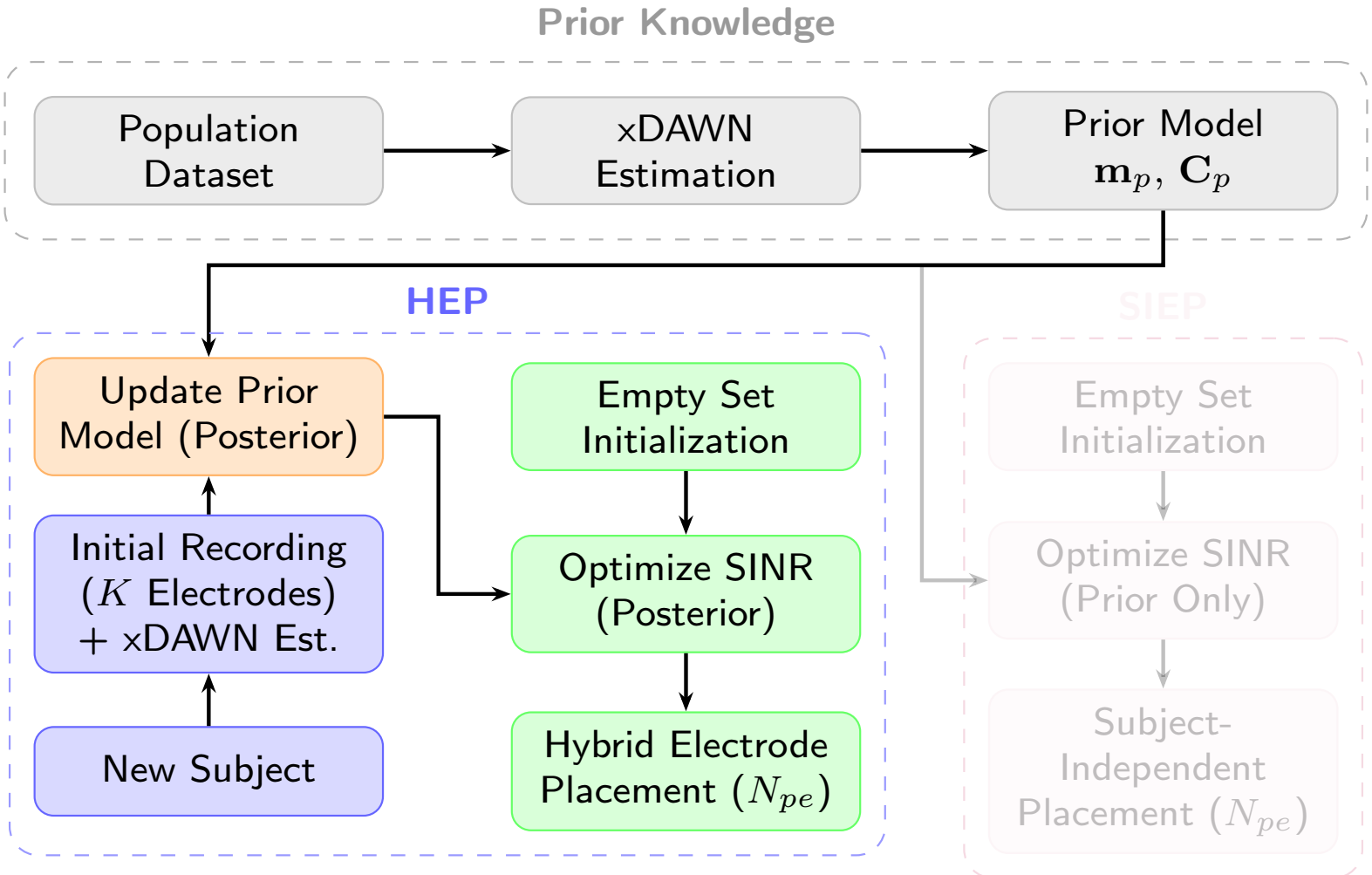
Hybrid Electrode Placement (HEP)

Combining Prior Knowledge with Subject-Specific Recordings

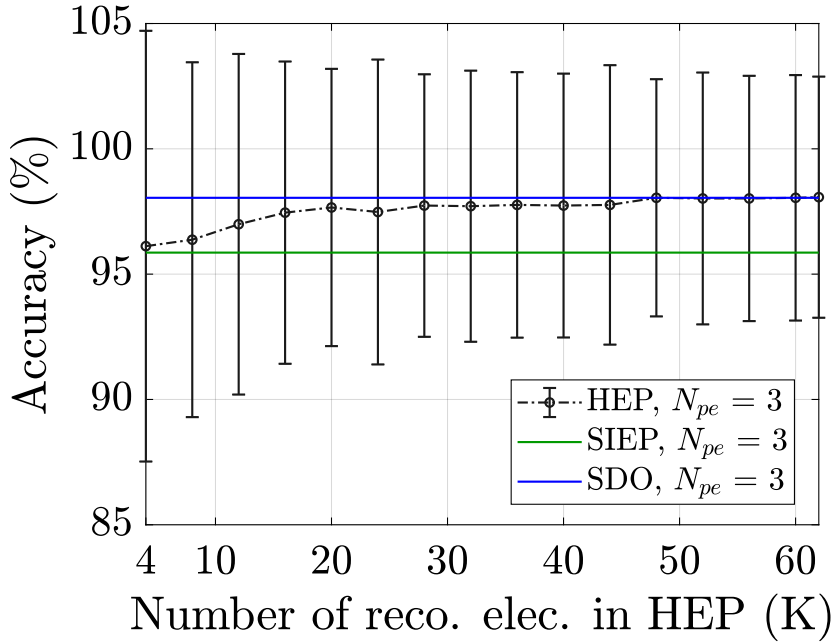


Hybrid Electrode Placement (HEP)

Combining Prior Knowledge with Subject-Specific Recordings



HEP results



- HEP accuracy grows from SIEP toward SDO as the number of initial recording electrodes (K) increases.

Introduction

Optimal Sensor Placement for Source Separation Under Noise

Application in P300-based brain computer interface

Sensor placement for sparse reconstruction

Conclusion

Underdetermined system with sparse sources

Same linear mixing model :

$$\mathbf{y}(\mathbf{X}_M, t) = \mathbf{A}(\mathbf{X}_M)\mathbf{s}(t) + \mathbf{n}(\mathbf{X}_M, t)$$

- **Setting :** $M < P$ — fewer sensors than sources
- **Challenge :** Linear source separation not possible in general, even with known $\mathbf{A}$
- **Solution :** Exploit sparsity : $\|\mathbf{s}(t)\|_0 = k \ll P \rightarrow$ sparse reconstruction algorithms

Sparse Reconstruction via BPDN

Recover $\mathbf{s}(t)$ by solving the **Basis Pursuit Denoising (BPDN)** :

$$\hat{\mathbf{s}}(t) = \arg\min_{\mathbf{s}} \|\mathbf{s}\|_1 \quad \text{s.t.} \quad \|\mathbf{y}(\mathbf{X}_M, t) - \mathbf{A}(\mathbf{X}_M)\mathbf{s}\|_2 \leq \epsilon$$

- ℓ_1-norm promotes **sparsity** as a convex surrogate for $\|\cdot\|_0$
- ϵ bounds the noise level

Successful reconstruction is guaranteed by properties of $\mathbf{A}(\mathbf{X}_M)$ such as :

- Mutual Coherence
- Restricted Isometry Property (RIP)

Proposed Sensor Placement Criteria

Mutual coherence-based criterion :

$$\hat{J}_{\mu} = \mathbb{E}\left[\left(\sum_{1\leq i\neq j\leq P}\left(\frac{\mathbf{a}_i^T\mathbf{a}_j}{\|\mathbf{a}_i\|_2\|\mathbf{a}_j\|_2}\right)^2\right)^{1/2}\right]$$

MSE-based criterion :

$$\hat{J}_{\mathrm{MSE}} = \mathbb{E}\left[\mathrm{Tr}\left(\mathbf{A}^T\left(\rho\mathbf{A}\mathbf{A}^T + \mathbf{C}^n\right)^{-1}\mathbf{A}\right)\right]$$

where $\rho = \frac{k}{P}$, measure of sparsity.

Experimental Setup

Sparse Reconstruction

- Domain $[0,1]^2$, $P = 30$ sources, $M = 25$ sensors $(M < P)$
- Sparse source vector with sparsity level $k = 1, 2, \ldots, M$
- Reconstruction via LASSO after noise whitening

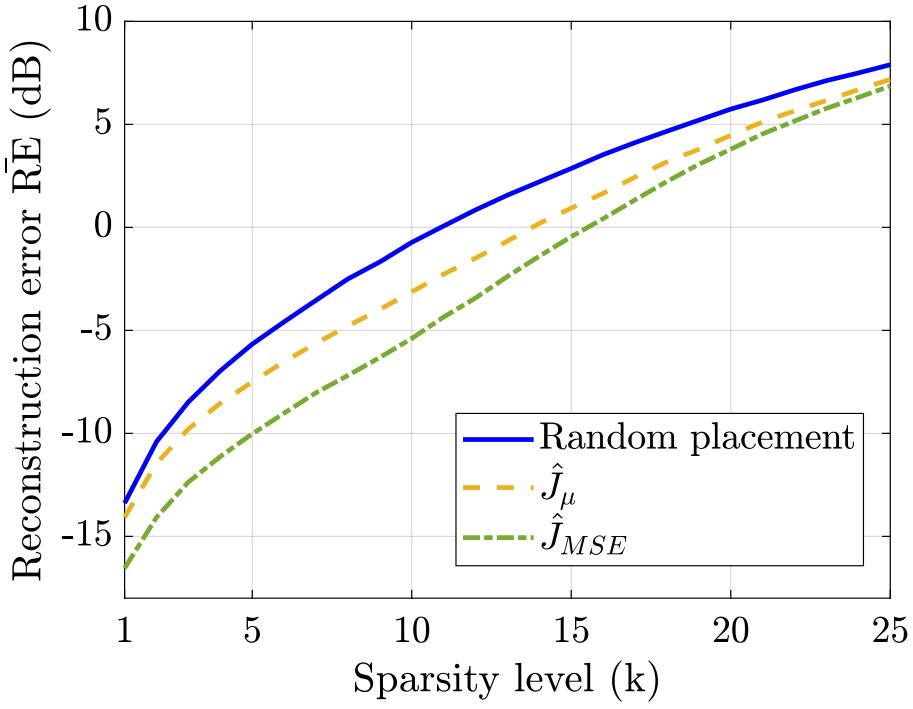
Placement Criteria :

- MSE-based criterion
- Mutual Coherence-based criterion
- Random placement

Metric : Average reconstruction error over 1500 signal samples and 20 MC runs.

Numerical result

Comparing MSE and Mutual Coherence Criteria



- MSE-based sensor placement : lower reconstruction error then mutual coherence based and random placement.

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Summary of Contributions

Optimal Sensor Placement for Source Separation with Noisy recordings (Sadeghi et al. Proc. EUSIPCO 2023, Sadeghi et al. Signal Processing 2025)

- Proposed a theoretical framework
  - Modeling uncertainty and prior knowledge with GP
  - Two criteria : SINR & MSE
  - Addressing optimization :
    - Efficient greedy multi-start gradient-based optimization
    - Sequential approach enables model update using gain estimation from already placed sensors (e.g., BSS estimation)
    - Sensor refinement
- Numerical simulations :
  - Outperforms entropy/MI and source-extraction criteria
  - Overestimating uncertainty level better than underestimating it

Summary of Contributions

Application to P300-based BCI (Sadeghi et al. BSPC 2026 (under review))

- Theoretical methods applied to a real-world application
  - **SIEP** : **prior**-knowledge-only based optimal sensor placement
  - **HEP** : **posterior**-knowledge based optimal sensor placement
- Validated on 3 datasets, with 2 classification methods
- Outperforms state of the art methods (SEP and RCES)
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OSP for Sparse Reconstruction with Noisy Recordings (Sadeghi et al. IEEE SPL (to be submitted))

- Proposed method extended to underdetermined case with sparse sources
  - Two criteria : MSE and mutual coherence
  - Efficient two stage gradient-based optimization
- Proposed MSE criterion outperforms classical mutual coherence criterion in noisy case

Perspectives

Short-term

- **Stochastic noise model**
  Apply GP-style uncertainty to noise covariance (currently deterministic)
- **Alternative optimizers**
  Evolutionary algorithms, reinforcement learning for global search

Long-term

- **Approximate expectation**
  Avoid costly Monte Carlo ; derive tractable bounds/approximation
- **Convolutive mixing**
  Extend to acoustic scenarios with propagation delays
- **Broader applications**
  Other EEG paradigms, biomedical signals, data-driven sensor selection

Thanks for your attention !

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